

CONTEXT-DEPENDENT PRECISION UTILITY FOR MIXED-PRECISION OBJECT DETECTION

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ABSTRACT

Mixed-precision quantization (MPQ) is a key enabler for deploying object detectors at the edge: assigning different bit-widths to different layers trades accuracy for cost far more efficiently than uniform quantization. The dominant recipe scores each layer’s precision sensitivity independently and allocates bits from these singleton scores. In this paper we show, through controlled measurements on Mask R-CNN with a Swin-T backbone, that this independence assumption breaks in detectors: precision utility is context-dependent, with pairwise interactions at $0.79\times$ the scale of singleton effects, 22.6–32.0% ranking reversals across five calibration seeds, and up to 3.6 AP of decision-level impact at equal budget. We then present a labels-free allocation method that measures conditional marginal utility directly on 64 unlabeled images, stabilizes the search with an ε -tolerant split-half criterion, and refines allocations with budget-neutral pairwise exchanges; a sample-complexity analysis explains why 64 images suffice. At matched parameter-weighted budgets on COCO, the method leads at every practical budget (e.g., +5.2 AP over uniform-4 at 4.0-bit and +1.8 at 4.5-bit) and transfers to a second backbone (Swin-S, +12.5 at 4.5-bit). Code is available at https://github.com/ldx825/Mixed-Precision-Object-Detection_ICASSP-2027.

Index Terms— mixed-precision quantization, object detection, bit allocation, calibration, model compression

1. INTRODUCTION

Deploying object detectors at the edge requires quantization [1, 2], and mixed-precision quantization (MPQ)—spending high precision only where it matters—promises a better accuracy–cost trade-off than uniform bit-widths. The dominant recipe is two-staged: (i) score the “sensitivity” of each parameter group (layer, block, or channel) independently, once, at full precision or at a fixed reference state; (ii) solve a budgeted allocation problem over these per-group scores [3, 4, 5, 6, 7, 8, 9]. A silent assumption underlies stage

(i): that a group’s contribution is an *intrinsic, context-free* property.

This paper shows that in object detection this assumption fails, and we build an allocation method around its failure.

(1) The problem: precision utility is context-dependent.

Using a controlled instrument—fake quantization of the 12 transformer blocks of a Swin-T backbone inside Mask R-CNN [10, 11], with a fully GT-free utility defined by agreement with the full-precision (FP) teacher detector on 64 unlabeled images—we measure precision utility in *context*: the marginal value of repairing a block given that other blocks are already repaired. Three findings emerge: pairwise interactions average $0.79\times$ the size of singleton effects; conditional and singleton rankings reverse on 22.6–32.0% of within-context pairs across five calibration seeds (Fig. 1); and at an identical repair budget, decisions made from singleton rankings lose 3.56 AP with respect to conditional measurement (singleton top-4: 41/495; conditional: 11/495, by exact enumeration).

(2) The method: GT-free conditional allocation.

We present a *conditional* allocation method: it estimates marginal utility in the current allocation context with the same instrument, stabilizes noisy 64-image estimates with an ε -tolerant split-half criterion over single upgrades and budget-neutral pairwise exchanges, and selects among several starting allocations by full-64 utility. It requires no labels, no retraining, and only $\sim 10^2$ – 10^3 calibration forwards.

(3) The results: best-or-tied at every budget, two backbones.

At matched parameter-weighted budgets on COCO 5k [12], the method attains the best-or-tied deployable accuracy at every tested budget (e.g., +1.8 over the strongest reimplemented prior at 4.5-bit), with the margins holding when the strongest baseline is granted the same extended bit-level set. It transfers to a second backbone of a different size (Swin-S, +12.5 at equal budget), and it comes with analyses that explain *why* it works: a sample-complexity account of why 64 calibration images suffice, and a reversal-floor law that predicts the measured reversal rate and yields a design criterion for when singleton scoring is structurally unreliable.

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2. RELATED WORK

Post-training quantization and bit allocation. PTQ has matured from uniform rounding [13, 14, 15] and learned step sizes [1, 16] to mixed precision, where layers receive different bit-widths under a size or bit budget [4, 3, 5, 17, 18], including reconstruction-based [19, 20] and activation-aware LLM objectives [21, 22]. Most mixed-precision methods follow the score-then-allocate template, which itself inherits the classical sensitivity-pruning lineage [23]: compute a per-layer sensitivity—often Hessian- or Fisher-based [3, 7], sometimes task-metric- or regression-based [9]—and then solve a budgeted program (DP, ILP, or Lagrangian search) over these independent scores.

Detector-specific quantization. Detector backbones have evolved from CNN detectors [24] to transformer detectors [25, 26] and ViT-style backbones [27], with dedicated PTQ designs [28, 29, 30]. F-LBQ [6] allocates transformer-layer bit-widths by minimizing backbone-feature alteration (uniform quantization on heads; layer-wise feature-space proxy). LampQ [7] uses a type-aware Fisher-style score with an ILP allocation and reports Mask R-CNN + Swin-T detection results at 4-bit weight/activation. Reg-PTQ [9] targets the regression branch with a global reconstruction loss. These works share the singleton-score premise; we instead measure *conditional* utility and show the singleton premise is quantitatively unreliable for detector allocation decisions.

Output-distortion objectives. BLOB-Q [8] approximates global distortion by a second-order expansion and, under a layer-uncorrelatedness assumption, decomposes it into additive terms enabling DP allocation; it reports Mask R-CNN + Swin-T/S detection at 4/5/6-bit. We test the *operational consequence* of additive decomposition: can context-free contributions predict which block deserves extra precision *after* others changed? Our interaction and rank-reversal measurements indicate they cannot (Sections 4, 7), which motivates measuring marginals in context.

Calibration without labels. Most quantization calibration uses unlabeled data [31, 32]; what varies is the quantity measured. Feature-space proxies [6], output-level quadratic objectives [8], and gradient/Fisher statistics [7] can all be computed without labels. Ours also needs no labels—only FP-teacher detections—but measures *task-output agreement in the current context* rather than a singleton proxy, with a sample-size analysis (Sec. 6).

Reimplementation note. Prior-art comparators are labeled reimplementations/stand-ins (feature-MSE for F-LBQ-style, additive output sensitivity for BLOB-Q-style, and a Fisher-style score for LampQ-style); they share our quantizer, calibration set and budget accounting, and are not the official pipelines.

3. PRELIMINARIES

Model and quantization contract. We use the official Mask R-CNN with Swin-T (mmdetection 3.3.0) and its released COCO checkpoint. FP bbox/mask AP is 42.65/39.28 on val2017 (5k). The allocation units are the 12 Swin-T blocks; each block comprises four weight groups {qkv, proj, fc1, fc2}, and every group receives symmetric min-max per-tensor weight-only fake quantization at a bit-width $b \in \{3, \dots, 8\}$ (the main sweep spans $\{4, \dots, 8\}$; the 4.0-bit regime additionally exploits level 3, see Sec. 7); heads, LayerNorm, embeddings and biases remain FP. Blocks differ $60\times$ in size (0.11–7.1M; Stage-3 dominates). The *budget* of an allocation $\mathbf{b}$ is the parameter-weighted average bit $\bar{b}(\mathbf{b}) = \sum_g n_g b_g / \sum_g n_g$, where n_g is the number of weights in group g . All methods compare at identical $\bar{b}$.

GT-free utility. Let the FP model be the teacher. For a quantized state with repaired set S (blocks in S at 8-bit, others at the base $\bar{b}$ configuration), run the standard test pipeline on a fixed unlabeled calibration set (64 COCO train images by default) and match teacher and student detections per image and class (greedy by teacher score, IoU ≥ 0.5 , teacher score ≥ 0.3). Define

$$U(S) = -\text{score_drop}(S) = -\overline{(s^{\text{FP}} - s^{\text{q}})}_{\text{matched pairs}}, \quad (1)$$

the negative mean score gap over matched teacher–student pairs (pooled over images; boxes only; no ground truth anywhere). The *conditional marginal utility* of repairing block i given context S is $\Delta(i | S) = U(S \cup \{i\}) - U(S)$, and the pairwise interaction is $I_{ij} = U(\{i, j\}) - U(\{i\}) - U(\{j\}) + U(\emptyset)$ (centered at the base state; without the $+U(\emptyset)$ term the raw interaction is dominated by the offset $|U(\emptyset)|$ and is uninformative). U is measured, not modeled: no Hessian, no proxy features, no labels.

4. DETECTOR PRECISION UTILITY IS CONTEXT-DEPENDENT

Non-additive interactions. We measured all $\binom{12}{2} = 66$ pairwise interactions on a 6-bit base (all blocks at 6-bit; $U(\emptyset)$ has score_drop 0.0298). Table 1 summarizes: the average absolute centered interaction is 0.00135 versus mean $|\Delta(i | \emptyset)| = 0.00171$, a ratio of 0.79. Extremes are instructive: BLK2.3+BLK2.5 are nearly useless alone yet interact by +0.0060, while two of the best singletons (BLK2.0/BLK2.2) underperform their sum by −0.0047.

Rank reversal. If interactions were small or homogeneous, singleton rankings would still be usable. They are not: comparing, for every context S , the conditional ranking $\Delta(i | S)$ against the singleton ranking $\Delta(i | \emptyset)$ over the remaining candidates, 22.6%, 24.7%, 26.4% of pairwise orders reverse at $|S|=1, 2, 3$ (Fig. 1; 12/66/220 contexts, top-1 mismatch 25/61/62%). Reversals are mostly moderate in

Table 1. Pairwise interactions, 6-bit base (calib-64, GT-free).

statistic	value
mean I_{ij}	+0.0005
std I_{ij}	0.0017
range	$[-0.0047, +0.0060]$
mean $ I_{ij} $	0.00135
mean $ \Delta(i \emptyset) $	0.00171
ratio	0.79

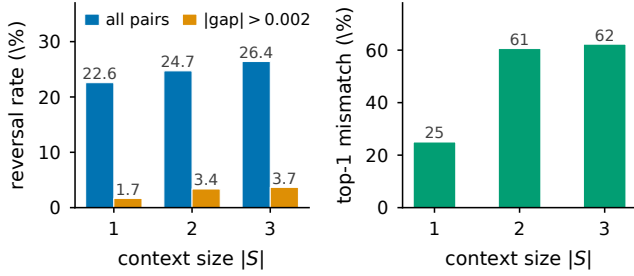


Fig. 1. Singleton-vs-conditional ranking under growing context: (a) pairwise order reversal rate (all pairs vs. large-gap pairs, $|\text{gap}| > 0.002$); (b) top-1 mismatch rate.

size, but the largest reversals change decisions: in context $S = \{\text{BLK2.5}, \text{BLK3.0}\}$ conditional measurement prefers BLK2.3 over the singleton-favored BLK2.4, a +0.0070 utility swing ($\approx 4 \times$ the mean singleton effect).

Five-seed replication. Four additional 64-image pools give reversal rates of 22.6–32.0% (all $\geq 20\%$) and interaction-to-singleton ratios of 0.50–1.12 (mean 0.76); per-pair values are seed-dependent, so we report distribution-level statements, which all replicate.

Decision impact. Utility-level differences translate to decisions. At an equal repair budget (4 of 12 blocks at 8-bit, 4-bit base), selecting blocks by singleton rankings of the same instrument gives 31.39 AP—about chance-level among repair sets (31.07)—while conditional measurement gives 34.95 (+3.56), and the oracle repair set gives 35.32. Exact enumeration of all $\binom{12}{4} = 495$ sets confirms the ordering (top-4: 41/495 vs. 11/495); predicted and measured gains correlate at Spearman 0.72–1.00 (legal signals) vs. -0.16 – 0.60 (feature proxy).

5. METHOD: GT-FREE CONDITIONAL ALLOCATION

The method has four components; Algorithm 1 summarizes the complete pipeline.

Conditional greedy. From the all-4-bit base, repeatedly perform the feasible single-level upgrade maximizing gain-per-cost $\Delta(i | S)/n_i$ (Eq. (1)), re-measuring in the new context after each commit; stop when no upgrade fits the residual budget. Scores are measured in the actual context in which the block will operate, unlike singleton scores computed once at

Algorithm 1 GT-free conditional allocation at budget $\bar{b}$.

Require: FP teacher; N blocks with precision ladder $\mathcal{L}$; budget $\bar{b}$; calibration set $\mathcal{C}$ (n unlabeled images); starts $\mathcal{S}_0$; tolerance ε

Ensure: deployed allocation $\mathbf{b}^*$

- 1: **Util**($\mathbf{b}$): run the test pipeline on $\mathcal{C}$ with $\mathbf{b}$; match teacher-student detections per image and class (greedy by score, at fixed matching thresholds); return $-\text{score_drop}$ (Eq. (1))
- 2: **Phase 1 (conditional greedy).** $\mathbf{b} \leftarrow \mathbf{b}_{\text{base}}$
- 3: **while** a feasible single-level upgrade fits the residual budget **do**
- 4: $i^* \leftarrow \arg \max_i [\text{Util}(\mathbf{b} + \mathbf{e}_i) - \text{Util}(\mathbf{b})]/n_i$ { $\mathbf{e}_i$ upgrades block i ; re-measured in context}
- 5: $\mathbf{b} \leftarrow \mathbf{b} + \mathbf{e}_{i^*}$; **append** $\mathbf{b}$ to $\mathcal{S}_0$
- 6: **end while**
- 7: **Phase 2 (ε -tolerant refinement), per start $\mathbf{b}_s \in \mathcal{S}_0$:**
- 8: **repeat**
- 9: split $\mathcal{C}$ into halves A, B ; for each move m in the neighborhood (single-level swaps; budget-neutral pair exchanges, one level up and one down) compute per-half score-gap changes $d_A(m), d_B(m)$
- 10: accept the best m with $d_A + d_B > 0$ and $\min(d_A, d_B) \geq -\varepsilon$ (Eq. (2)); $\mathbf{b}_s \leftarrow \mathbf{b}_s + m$
- 11: **until** no move accepted
- 12: **Phase 3 (selection).** **return** $\mathbf{b}^* \leftarrow \arg \max_{\mathbf{b}_s \in \mathcal{S}_0} \text{Util}(\mathbf{b}_s)$ {full-64 utility; no labels, no 5k}

a fixed reference.

ε -tolerant refinement. 64-image measurements are noisy (cross-seed std 0.0016, comparable to typical swap effects, Sec. 6). We split the 64 images into two halves and accept a move iff

$$d_A + d_B > 0 \quad \text{and} \quad \min(d_A, d_B) \geq -\varepsilon, \quad \varepsilon = 10^{-3}, \quad (2)$$

over a neighborhood of single-block ± 1 swaps and budget-neutral pairwise exchanges ($i: +1, j: -1$) at constant $\bar{b}$; iterate to a fixed point.

Multi-start and cost. We select among several starts (conditional greedy plus strong baseline-produced allocations) by full-64 utility—no ground truth and no 5k data enter the search. Per budget the pipeline uses $\sim 10^3$ – 10^4 detector forwards (~ 400 greedy, ~ 600 – 900

6. SAMPLE-COMPLEXITY ANALYSIS

The method ranks candidates and accepts moves by comparing estimated utilities $\hat{\mu}_a = n^{-1} \sum_k Z_a(x_k)$, where $Z_a(x)$ is the bounded per-image utility contribution of move a ($|Z_a| \leq 1$). Two standard consequences link the calibration size n to allocation reliability.

(T1) Correct comparisons. Let there be M candidate moves per round and R rounds ($M \leq 150$, $R \leq 8$ in our setting). By Hoeffding’s inequality [33] and a union bound, if the comparison margin between the best and second-best move is $\Delta\mu$ (in per-image mean units), then

$$n \geq \frac{2}{\Delta\mu^2} \log\left(\frac{2MR}{\delta}\right), \quad (3)$$

suffices for the measured search trajectory to match the exact-utility trajectory with probability $\geq 1 - \delta$.

(T2) ε -optimal acceptance. The acceptance rule Eq. (2) only asks for moves that are ε -close to optimal; the standard relaxation gives

$$n \geq \frac{2}{\varepsilon^2} \log\left(\frac{MR}{\delta}\right), \quad (4)$$

so tolerance and sample size trade off quadratically—the methodological reason a small ε paired with 64 images is a principled operating point rather than a heuristic.

(T3) Interaction-conditioned reversal floor. Two candidates a, b flip order between singleton and context- S evaluation iff the interaction sum $X_S = \sum_{j \in S} (I_{aj} - I_{bj})$ opposes and exceeds the singleton gap $D = \Delta_a(\emptyset) - \Delta_b(\emptyset)$. Under symmetric independent errors the expected reversal rate is

$$R = \frac{1}{2} \mathbb{P}(|X_S| > |D|). \quad (5)$$

On our measured tables Eq. (5) predicts $R = 22.1\%$ at $|S|=1$; exact enumeration with the measured U values gives 22.58%, equal to the independent Exp31 measurement. Design criterion: $R < 5\%$ needs $\rho = |I|/|\Delta|$ below 0.134—detectors measure 0.76: no calibration budget repairs a singleton design.

Theory meets measurement. At $n=64$ the cross-seed noise (0.0016) is $0.75\times$ the mean effect (0.0021) and shrinks as $1/\sqrt{n}$; top-1 selection stabilizes at 64–128 images, matching the $(1/\Delta\mu^2)$ scaling of Eq. (3).

Scope. The analysis is at the level of the *proxy utility* actually measured (proxy-to-AP is empirical, Spearman 0.72–1.00); the guarantee is that noise does not misdirect the local search—not global optimality.

7. EXPERIMENTS

Setup. Full COCO val2017 (5k) bbox AP with pycocotools under the Sec. 3 contract. Baselines: **uniform**; **F-LBQ-reimpl** (feature-MSE greedy); **BLOBQ-reimpl** (output-damage curves); **LampQ-style** (Fisher-style); **static-OUT**; **oracle-approx** (reference).

Main results. Our method attains the best-or-tied deployable AP at every tested budget: +5.24 over uniform-4 at 4.0-bit (extended level set; ties the strongest fair-contract baseline at 22.65), +1.77 at 4.5, +0.04 at 5.0, +0.44 at 5.5, and +0.05 at 6.0 (a completion step spends the leftover 0.05 bit). The

Table 2. COCO 5k bbox AP at matched parameter-weighted budgets ($\bar{b}$); bold = best deployable per budget. FP = 42.65. F-LBQ/BLOBQ/LampQ are reimplementations; s-OUT = static-OUT. “Ours” uses the extended level set (level 3); under the same extended contract the strongest baseline also reaches 22.65 at 4.0-bit and 35.72 at 4.5-bit (cf. fairness notes).

$\bar{b}$	unif.	F-LBQ	BLOBQ	LampQ	s-OUT	Ours
4.0	17.41	17.41	17.41	17.41	17.41	22.65
4.5	17.41	35.03	34.74	32.83	33.84	36.80
5.0	39.57	37.93	39.87	39.57	39.57	39.91
5.5	39.57	40.28	40.92	40.94	41.18	41.62
6.0	41.96	41.84	41.94	41.96	41.96	42.01

cross-strategy spread is 3.9 AP at 4.5-bit (32.83–36.80), collapsing to 0.12 AP at 6.0-bit.

What wins. Winning allocations pile precision on *shallow, cheap* blocks (stages 0–1, 0.1–0.4M weights; up to 8-bit) while the dominating stage-3 blocks receive *differentiated* precision: at 4.5-bit one parameter-neutral Stage-3 swap ($4 \rightarrow 3$ next to $4 \rightarrow 5$) is worth +1.21 AP. Reallocation—not singleton ranking—separates the methods: uniform-4 scores 17.41 AP, while the same budget reallocated reaches 36.80.

Ablation: search design. At 4.5-bit: strict split-half validation rejects true improvements (full-chain: 33.84, -1.75); the ε -tolerant criterion plus pair exchanges (incl. level-3 moves) reaches 36.80.

Second backbone. On Mask R-CNN + Swin-S (24 blocks, same contract; FP = 48.18), the same procedure reaches 44.74 AP at the 4.5-bit budget versus 32.22 for uniform-4—not Swin-T specific.

Calibration-size behavior. Five 64-image pools give cross-seed top-1 agreement of 80%/100% at $n=64/n \geq 128$ (Eq. (3)).

Fairness notes. All comparators share our quantizer (weight-only) and budget; prior published numbers use weights and activations together and are therefore not directly comparable to ours (bridging the two is follow-up work). Under the fair extended contract (level 3 granted to all), our deployed outputs lead at 4.5–6.0 (+1.08, +0.04, +0.44, +0.05) and tie the strongest baseline at 4.0 (22.65); the feature-MSE baseline collapses to 25.48 at 4.5, exposing the 3-bit blind spot of singleton signals.

8. CONCLUSION

Singleton scoring assumes each layer’s precision sensitivity is fixed. On a Mask R-CNN + Swin-T detector it breaks: precision utility is context-dependent—pairwise interactions match singleton effects ($0.79\times$), 22.6–32.0% of within-context rankings reverse, and the errors cost measurable AP (-3.56). We address the failure with a GT-free method that measures conditional marginal utility on unlabeled images, using an ε -tolerant split-half search and budget-neutral exchanges; it leads at every tested budget and transfers to Swin-S. Limitations: weight-only, single family, proxy theory; future work: weight-and-activation alignment and broader detector families.

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